

Ashkenazi Population Structure Reassessed: Reference-Panel Sensitivity and the Europe–Levant Cline

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Abstract

Population-genetic inference is conditional on reference-panel design, yet the sensitivity of Ashkenazi Jewish ancestry models to central Mediterranean comparator omission has rarely been tested directly. This study combines author-generated qpAdm and autosomal FST analyses with a linked fifteen-experiment Global25 sensitivity suite and a critical synthesis of published PCA, identity-by-descent, uniparental, ancient-DNA, and historical evidence. In Global25 Experiment 10, the full modern panel identifies Italki as the largest individual source (40.8) and assigns 54.6 to the strict Italian Jewish and central Mediterranean block at distance 0.00819655. Removing only Italki worsens fit by 12.49% and reallocates most of its coefficient to related Italian and central Mediterranean sources. Removing Italki together with Southern Italian, Sicilian, Maltese, and Aegean comparators worsens fit by 47.18%, raises Lombardy from 9.8 to 38.0, and raises the Levantine-continuity block from 22.8 to 52.0. Reciprocal Italki models, ten regional Ashkenazi targets, prehistoric controls, and ancient Levant positive controls reproduce the expected directional behavior. These results show that omission of central Mediterranean comparators can generate or exaggerate an apparent Europe-Levant cline by forcing the target to be represented by the remaining north-shifted Italian and Levantine poles. Global25 coefficients are descriptive, panel-dependent similarity weights rather than literal ancestry percentages. A full-dimensional diagnostic of the German regional model further shows that its Levantine-continuity and Northern/Eastern European blocks generate strongly opposing, nearly equal weighted displacements around a dominant central Mediterranean reconstruction. The inference rests on matched redistribution, fit deterioration, reciprocal controls, vector geometry, regional replication, positive controls, and convergence with independent evidence. Within those limits, the results support a primary Italian Jewish and central Mediterranean affinity, while the smaller proxy coefficients require geometric rather than literal additive interpretation.

Keywords: Ashkenazi Jews; population genetics; reference-panel bias; comparator omission; central Mediterranean; Southern Italy; qpAdm; FST; Global25; sensitivity analysis

Part I. Data Sources, Methods, and Analytical Framework

qpAdm Modeling

I conducted the author-generated qpAdm analyses in the IllustrativeDNA Admix Lab web interface using the Human Origins v62.0 dataset. The target was Jew_Ashkenazi.HO, and the primary three-way source configuration was Italian_South.HO, Lebanese.HO, and Russian.DG. I also evaluated alternative specifications that replaced Italian_South.HO with Northern or Central Italian sources. The interface was run with allsnps=TRUE. The exact source labels, right-population set, displayed coefficients, fit diagnostics, screenshots, and copied outputs are retained in the reproducibility repository. Because these analyses were executed through a web implementation rather than a local ADMIXTOOLS command-line workflow, they are treated as comparative validation analyses rather than as a substitute for fully re-estimated raw-genotype models.

I interpret qpAdm models comparatively, prioritizing feasibility, fit, and stability across historically plausible source configurations. The central question is whether inclusion or exclusion of Southern Italian and central Mediterranean references changes model behavior and the partitioning of Ashkenazi affinity. No single feasible model is treated as uniquely identifying historical populations, because closely related Mediterranean sources can substitute for one another and modern proxies are not identical to ancestral populations.

Pairwise FST Analysis

I obtained pairwise autosomal FST distances through the same IllustrativeDNA Admix Lab environment, with Adjust Pseudohaploids set to NO. Comparisons included Southern Italian, Sicilian, Maltese, Greek and Aegean, Levantine, and additional European populations. FST is used descriptively to rank relative affinities and is calibrated against within-Europe contrasts, including north–south Italian and internal Iberian distances. Standard errors are reported where supplied by the interface, but no formal significance tests are inferred from the pairwise ranking. These results support comparative ordering, not precise demographic parameter estimates.

Global25 Sensitivity and Validation Suite

I evaluated Global25 results through a linked suite of fifteen experiments rather than through isolated centroid distances or two-dimensional plots. Experiments 1 and 8 use fixed prehistoric source panels; Experiments 2 and 10 use broad modern competition panels; Experiments 3, 5, 11, and 12 are controlled omission tests; Experiments 4, 6, and 13 provide direct-distance and reciprocal Italki controls; Experiments 7 and 14 test ten regional Ashkenazi population averages; and Experiments 9 and 15 test whether the framework recovers genuine ancient Levantine continuity. Prehistoric and modern sources are analyzed in separate panels. The principal evidence is the direction and magnitude of matched redistribution, deterioration of fit after omission, reciprocal behavior, regional persistence, and positive-control performance.

Experimental Settings and Interpretation

All experiments use scaled Global25 coordinates and declared population-average targets and sources. Runs were recorded from Thierry Kerneur's ExploreYourDNA Atlantic Bronze Age calculator with cycles set to 0.25X, distance column off, printed zeroes off, fast mode off, and aggregate mode on. Source and target labels, raw outputs, grouped summaries, screenshots, comparison files, and checksums are retained in the repository. Displayed coefficients are nonnegative, constrained, source-panel-dependent similarity weights; they are not exact historical ancestry percentages and have no formal confidence intervals. Correlated sources can substitute for one another, and an unprinted or zero coefficient does not prove historical absence. Regional averages with $n=2$, ancient averages with $n=3$ or $n=4$, and the Italki average with $n=9$ require particular caution.

Full-Dimensional Vector Diagnostic

To clarify how category-level coefficients relate to the target's position, I performed a full twenty-five-dimensional vector diagnostic for the Ashkenazi German target in Experiment 14. The sources retained in its displayed solution were grouped as strict central Mediterranean (Reggio Calabria, Italki, and Maltese), Levantine continuity (Lebanese Arab Christian and Palestinian Arab Christian), and Northern/Eastern European (German and Polish). Within each block, a coefficient-weighted centroid was calculated after normalizing the retained source weights to the block total. Vectors were then drawn from the central Mediterranean centroid to the Levantine and Northern/Eastern European centroids. Their cosine similarity measures directional opposition, while each vector's norm multiplied by its displayed block coefficient measures its contribution to the reconstructed displacement. The primary calculations use all twenty-five Global25 dimensions. A PCA of the combined Experiment 14 source and target coordinates is used only to visualize the same geometry; PCA-axis signs are arbitrary and no ancestry proportion is inferred from the two-dimensional projection.

Original Analyses and Derived Data Generated in This Study

This study generates original quantitative outputs from alternative qpAdm source configurations, targeted autosomal FST comparisons, and the fifteen-experiment Global25 sensitivity and validation suite. The underlying coordinate and genotype data derive from previously published or publicly distributed datasets; the panel construction, matched comparisons, grouped summaries, figures, and interpretation were generated for this study. Supplementary Tables S1-S10 document the Y-chromosomal, qpAdm, FST, and calibration analyses, while the repository provides the Global25 source and target files, raw outputs, derived summaries, screenshots, calculation scripts, methods notes, and SHA-256 checksums.

Interpretive Framework

Genome-wide autosomal structure provides the primary basis for population-level affinity and ancestry inference because it reflects contributions across the genome. Uniparental markers represent single lineages and can be strongly affected by drift, founder effects, and endogamy. I therefore treat autosomal results as the primary comparative evidence and use Y-chromosomal results only as a consistency check. Neither broad haplogroup labels nor the frequency of a single lineage is treated as a population-wide ancestry estimate.

Reference populations are modeling assumptions rather than neutral inputs. Northern Italian, Tuscan, and Sardinian populations are therefore evaluated as specific proxies, not as interchangeable representatives of all Italian variation. The central hypothesis is that omitting Southern Italian, Sicilian, Maltese, Italian Jewish, and Aegean comparators removes the population space most relevant to the target. In a constrained model, the remaining north-shifted Italian and Levantine sources can then bracket the omitted space and create an apparent cline even when neither proxy is the best direct comparator.

Statistical and Inferential Scope

The fifteen Global25 experiments form one linked sensitivity analysis, not fifteen independent hypothesis tests. No p-values, confidence intervals, admixture dates, or multiple-testing claims are derived from the calculator outputs. The strongest inference is therefore the repeated directional response to controlled panel changes, especially the matched Experiments 10-12, supported by reciprocal, regional, and positive controls. Historical labels assigned to modern or ancient proxies should be read as geographic shorthand for sampled genetic variation, not as proof that the named populations were literal ancestral sources.

Rationale for Reanalysis Without New Sequencing

This study introduces no new biological sampling or sequencing because its primary question concerns reference construction and proxy choice. Reanalyzing existing data under explicit alternative panels is appropriate for testing whether an inferred Europe-Levant cline is stable or instead changes when relevant central Mediterranean comparators are restored. The analyses do not attempt to replace future raw-genotype or ancient-DNA work; they identify panel assumptions that such work should test directly.

Reproducibility and Repository Access

The complete reproducibility package for the author-generated qpAdm, FST, and Global25 analyses is publicly available at <https://github.com/StevenParker-collab/ashkenazi-dna-revised-pci-reproducibility>. The frozen archival release analyzed here is available at <https://doi.org/10.5281/zenodo.21383280>. It contains the declared inputs, exact labels and settings, raw and grouped outputs, scripts, figures, screenshots, methods notes, README documentation, and SHA-256 checksums for Experiments 1-15. The files are organized to permit independent inspection of each matched comparison and to distinguish source data from author-generated derivatives.

Part II. Reference Population Design and the Appearance of Genetic Intermediacy

Ashkenazi Jews are frequently described as genetically intermediate between Europe and the Levant. That description may reflect demographic history, but it can also arise when the reference panel represents Europe mainly with Northern or Central European groups, Northern Italians, Tuscans, or Sardinians while omitting Southern Italian, Sicilian, Maltese, Italian Jewish, and Aegean comparators. If the target lies near the omitted central Mediterranean space, the available north-shifted European and Levantine references can become the two nearest remaining directions. The target will then appear clinal by construction. Here, intermediacy refers to placement relative to a declared reference set, not to a reference-free biological property.

A central example is the neighbor-joining tree presented by Ostrer and Skorecki (2013), which has been widely cited as a summary of autosomal evidence for shared ancestry among Jewish Diaspora populations together with variation in admixture with surrounding non-Jewish populations.

Fig. 2 Neighbor-joining tree showing the relationship of European, Jewish, Middle Eastern, and North African populations, using F_{st} as the distance metric. The neighbor-joining algorithm used F_{st} as the distance metric input for calculation of a matrix specifying the distance between each pair of groups and then iterates until the tree is resolved and branch lengths discerned. The tree was rooted using the reference mixed Central and Southern African population as an out-group. Major population groups are labeled at the right

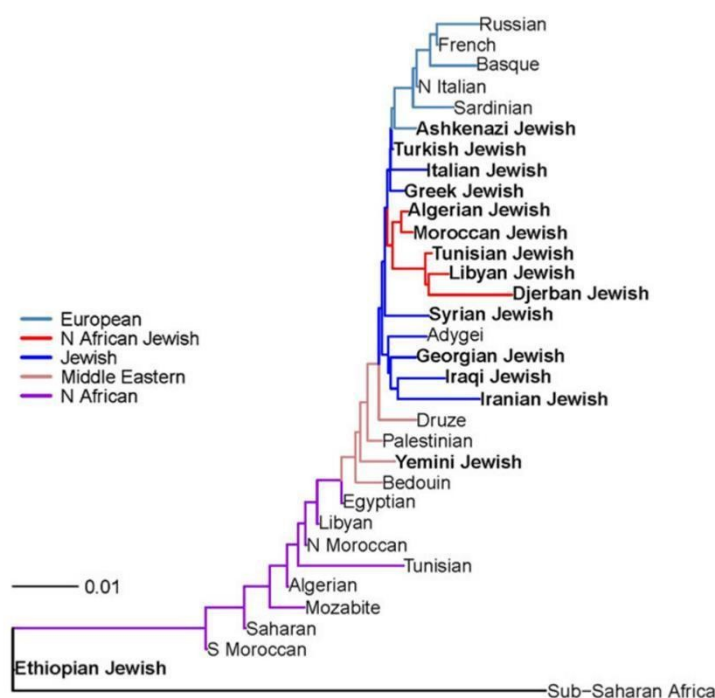


Figure 1. Neighbor-joining tree reproduced from Ostrer and Skorecki (2013). Tree constructed from autosomal genetic distances using Northern Italians, Tuscans, Sardinians, Basques, French, Russians, and other Northern or Central European populations as European references. Southern Italian and Sicilian populations are not included.

Ostrer and Skorecki describe their synthesis as drawing on a common set of autosomal studies, including Atzmon et al. (2010), Behar et al. (2010), Kopelman et al. (2009), Bray et al. (2010), Listman et al. (2010), and Campbell et al. (2012). They emphasize that these studies differed in the populations sampled and sample sizes, yet report broadly similar conclusions, which they interpret as evidence for shared genetic ancestry among major Jewish groups together with varying degrees of admixture from local host populations.

The studies cited in this synthesis share a consequential design limitation: Southern Italian and Sicilian autosomal comparators are absent. Europe is commonly represented by Northern Italians, Tuscans, Sardinians, and Northern or Central European groups. Sardinians are informative in their own right but, because of island-specific drift and distinctive demographic history, are not neutral substitutes for mainland Italian variation. Including Italian Jews also does not replace non-Jewish Southern Italian and Sicilian comparators when the question is whether Ashkenazi Jews fall within a wider regional continuum.

This asymmetry limits what the topology can establish. When Southern Italian and Sicilian variation is not represented, Ashkenazi samples cannot be evaluated for placement within that missing space. An apparent position between a north-shifted European pole and Levantine populations may therefore combine genuine population history with geometry created by the included comparators. The model cannot distinguish those possibilities without a sensitivity analysis that restores the omitted populations.

Accordingly, similar conclusions across studies are not fully independent when the studies reuse a similar reference scaffold. Repetition under the same omission can reproduce the same cline. This does not invalidate the underlying datasets; it means that the geographic interpretation must be retested with Southern Italian and central Mediterranean comparators included directly.

This reference template also shaped interpretation when proximity to Italian populations was observed. For example, Bray et al. (2010) note proximity of Ashkenazi Jews to Italian samples but propose that admixture with Central and Eastern European host populations could shift Ashkenazi Jews toward Italians along principal component axes relative to Middle Eastern populations. This hypothesis was advanced without directly testing Southern Italian and Sicilian ancestry within the same framework, despite its historical plausibility and relevance to interpreting Italian proximity. More generally, principal component geometry can be sensitive to reference population choice and does not by itself identify the historical sources responsible for intermediate placement.

I therefore treat the Europe–Levant cline as a testable, panel-dependent hypothesis. The key prediction is directional: restoring relevant central Mediterranean comparators should improve fit and absorb signal from both the north-shifted Italian and Levantine proxies; removing them should worsen fit and inflate those proxies. Experiments 10–12 test that prediction under matched settings. *Summary of Shared Reference Population Design Features (Descriptive Synthesis)*

Summary Table. Shared Reference Design Features Across Cited Studies

Category	Populations included across cited studies	Relevant populations excluded	Structural consequence / expected effect on placement
 Italian reference populations	✓ Northern Italians; Tuscans; Sardinians	✗ Southern Italians; Sicilians	→ Italy modeled northward; southern Italian structure absent
 European pole definition	✓ Northern and Central Europeans combined with Northern Italians and Tuscans	✗ Mediterranean Southern Europeans	→ Europe represented northward
 Italian proxy practice	✓ Sardinians used as an Italian proxy; Northern and Central Italians treated as representative of Italy	✗ Southern Italian population structure excluded; Italy modeled northward	→ Southern European continuum incompletely represented
 Jewish populations included	✓ Ashkenazi; Sephardi; Mizrahi; Italian Jews (included in some analyses and in the Ostrer and Skorecki synthesis)	⊖ Not applicable at the level of Jewish group inclusion; the primary omission concerns non-Jewish Southern Italian and Sicilian comparator populations	→ Jewish groups included, but the key omission is the absence of non-Jewish Southern Italian and Sicilian references
 Mediterranean comparators	✓ Limited or none among non-Jewish central Mediterranean comparators	✗ Southern Italy; Sicily; Maltese; Aegean Greeks	→ Mediterranean context inadequately represented
 Overall structural consequence	⊖ Europe approximated using northern proxies; Mediterranean structure only partially captured	✗ Southern European continuum incompletely represented	→ Apparent Europe–Levant intermediacy
 Expected effect on placement	⊖ Apparent Europe–Levant intermediacy	✗ Mediterranean anchoring cannot be evaluated	→ Mediterranean anchoring cannot be evaluated

The literature review identifies a recurring comparator gap; the matched experiments show its measurable effect. With the full panel in Experiment 10, the strict Italian Jewish and central Mediterranean block is 54.6 and distance is 0.00819655. Removing Italki alone worsens distance by 12.49% and transfers most of the missing coefficient to related Italian and central Mediterranean sources. Removing the wider central Mediterranean source space worsens distance by 47.18%, raises Lombardy from 9.8 to 38.0, and raises the Levantine-continuity block from 22.8 to 52.0. Within this declared panel, the Northern Italian–Levantine cline is therefore not stable: it emerges as the two remaining proxy directions absorb omitted central Mediterranean signal.

Part III. Y-Chromosomal Structure of Ashkenazi Jews in Mediterranean Context

Limits of Geographic Inference from Y-Chromosomal Data

Several early Y-chromosomal studies were interpreted as evidence for predominantly Levantine paternal ancestry, principally because haplogroups J1, J2, and E were labeled Near Eastern. Such geographic assignment requires caution. Broad haplogroups span multiple Mediterranean regions, and their frequencies can be strongly altered by founder effects and drift. Low phylogenetic resolution, pooled Jewish samples, and absent Southern European comparators therefore limit what these datasets can establish about Ashkenazi-specific paternal origins.

With the exception of Semino et al. (2004), the major Ashkenazi Y-DNA analyses from this period did not include Southern Italian populations as comparators, despite their central relevance to Mediterranean genetic structure and their well-documented Jewish presence since antiquity. Instead, these studies typically contrasted Ashkenazi Jews with Middle Eastern populations on one side and Northern or Central European populations on the other. This reproduces, in uniparental form, the same asymmetric reference framework later observed in autosomal studies. Under such a design, haplogroups shared broadly across the eastern Mediterranean and Southern Europe are implicitly reclassified as Levantine by default, not because their geographic specificity has been demonstrated, but because Southern European baselines are absent.

Hammer et al. (2009) illustrates the subgroup problem. The study pools Ashkenazi and non-Ashkenazi Jewish samples for several summaries and does not provide a complete Ashkenazi-specific population frequency table. Those pooled distributions should not be cited as if they were Ashkenazi-specific estimates. Coarse labels such as J or E-M35 also obscure subclade-level

geographic structure and can make widely distributed Mediterranean lineages appear more regionally diagnostic than the data support.

Other frequently cited studies show similar limitations. Behar et al. (2004) and Nebel et al. (2001) report Ashkenazi Y-chromosomal frequencies using broad haplogroup bins and without Southern European comparators. Shen et al. (2004) uses a small Ashkenazi sample and was not designed to distinguish Levantine-specific from broadly Mediterranean lineages. J2 and E-M35 are common across Southern Italy, Greece, Anatolia, and parts of the Levant; their presence alone cannot locate a paternal source.

Semino et al. (2004) is informative because it includes multiple Southern Italian populations and reports greater downstream resolution for haplogroups E and J. In that comparison, the apparent geographic specificity of the Ashkenazi signal is substantially reduced: Ashkenazi frequencies of J2 and E-M123 fall within Southern Italian ranges, whereas several Levantine populations display different relative frequencies, including higher J1 and lower E-M123. This overlap does not prove a uniquely Southern Italian paternal origin, but it shows that these lineages cannot be assigned to the Levant by default.

The early Y-chromosomal literature therefore does not independently establish a predominantly Levantine Ashkenazi paternal origin. When Southern Italian and Aegean comparators are included, the observed lineage frequencies are compatible with a broader Mediterranean distribution shaped by founder effects and endogamy. Because a uniparental marker represents only one lineage, this evidence is treated as contextual and cannot override genome-wide autosomal analyses.

As summarized in Table 1, the early Y-chromosomal literature does not provide consistent or properly contextualized evidence for a predominantly Levantine paternal origin of Ashkenazi Jews. Resolving this ambiguity requires focusing on the only dataset from this period that places Ashkenazi Jews and Southern Italian populations side by side at subclade resolution.

Table 1. Y-DNA haplogroup frequencies among Ashkenazi Jews (E, J1, J2 only).

Study	Ashkenazi Study sample size	E-E total	E-M78	E-M123	J1-M267	J2-M172	Notes
Hammer et al. 2009	1,575 (pooled)	NR	~3	~17	~17	~20	Ashkenazi and non-Ashkenazi Jews pooled; subgroup frequencies not reported
Behar et al. 2004	~442	~16.1	NR	NR	~19	~19	No Southern European comparators; coarse resolution
Semino et al. 2004	77	NR	5.2	11.7	14.6	23.2	Only study including Southern Italians; higher subclade resolution
Nebel et al. 2001	~79	23	NR	NR	~19	~24	No Southern European comparators
Shen et al. 2004	~20	NR	~10	~10	~20	~15	Very small sample; limited resolution

Table 1 note. In all tables in this manuscript, missing source values are coded as NR (not reported in the cited study). Values prefixed by ~ are approximate as reported. All haplogroup frequencies are percent.

Reassessment of Ashkenazi Y-Chromosomal Structure Using Southern Italian Comparators

The only early Y-chromosomal study to directly compare Ashkenazi Jews with Southern Italian regional populations at subclade resolution is Semino et al. (2004). Unlike later syntheses that rely on pooled Jewish samples or coarse haplogroup bins, this study reports haplogroups E and J for Ashkenazi Jews alongside multiple Southern Italian regional samples, including Apulia, Calabria, and Sicily. This design enables direct evaluation of whether Ashkenazi paternal lineages commonly labeled as Levantine are exceptional relative to Southern Europe.

At this resolution, Ashkenazi Y-chromosomal structure is dominated by haplogroups E and J2, both of which fall within Southern Italian variation. Ashkenazi Jews show E-M123 at 11.7 percent and E-M78 at 5.2 percent. These values closely match Southern Italian regions such as Apulia (E-M123 = 11.6 percent), Calabria 2 (E-M123 = 13.2 percent), and Sicily (E-M78 = 12.7 percent). Calabria 1 shows the inverse pattern, with high E-M78 (16.3 percent) and low E-M123 (2.5 percent), underscoring strong regional heterogeneity within Southern Italy itself. This heterogeneity shows that neither E-M123 nor E-M78 is geographically diagnostic of the Levant.

Haplogroup J shows the same pattern of Mediterranean continuity. In Semino et al. (2004), Ashkenazi Jews exhibit J2 (J-M172) at 23.2 percent, making it the dominant J lineage. Comparable or higher J2 frequencies are observed in Apulia (29.1 percent), Calabria 1 (22.8 percent), and Calabria 2 (20.0 percent), while Sicily shows moderately lower J2 (16.7 percent). In both Ashkenazim and Southern Italians, J2 exceeds J1, a structure that contrasts sharply with Levantine populations.

J1 (J-M267) is present in Ashkenazi Jews at 14.6 percent, elevated relative to some Southern Italian regions but well within a Mediterranean range. Southern Italy itself exhibits substantial regional variation in J1, ranging from very low values in Apulia (2.3 percent) and Calabria 1 (1.8 percent) to higher values in Sicily (7.1 percent). This intra-regional spread shows that J1 is not absent from Southern Europe and that moderate J1 frequencies cannot be treated as evidence of Levantine origin in isolation.

Crucially, the Ashkenazi J1 frequency remains far below Levantine paternal structures, in which J1 typically dominates the Y-chromosomal pool and often exceeds J2. The Ashkenazi pattern instead mirrors Southern European structure, with J2 dominant and J1 secondary. Given the long-documented founder effects and endogamy in Ashkenazi populations, modest elevation of J1 relative to some Southern Italian regions is parsimoniously explained by genetic drift acting on a small number of founding lineages rather than by descent from a Levantine paternal majority. When Southern Italian comparators are included, the supposed Levantine signal in Ashkenazi Y-DNA disappears. Ashkenazi Jews align with a Mediterranean continuum characterized by substantial E-M123 and J2, moderate and variable J1, and strong regional heterogeneity rather than with the J1-dominant structure typical of Levantine populations. The interpretation of Ashkenazi paternal ancestry as predominantly Levantine arises primarily when Southern Europe is excluded and when drift-sensitive lineages are over-interpreted without comparative context. Table 2 summarizes the Semino et al. (2004) data directly.

Table 2. Semino et al. (2004) Y-DNA haplogroup frequencies for Ashkenazi Jews and Southern Italian regions (E, J1, J2 only).

Population	Sample size	E-M78	E-M123	J1-M267	J2-M172
Ashkenazi Jews	77	5.2	11.7	14.6	23.2
Italian (Apulia)	86	0.0	11.6	2.3	29.1
Italian (Calabria 1)	80	16.3	2.5	1.8	22.8
Italian (Calabria 2, Albanian community, Cosenza)	68	5.9	13.2	0.0	20.0
Italian (Sicily)	55	12.7	3.6	7.1	16.7

In contrast, Levantine populations are characterized by a fundamentally different Y-chromosomal structure, with J1 typically exceeding J2 and E-M123 occurring at substantially lower frequencies than those observed in both Ashkenazi Jews and Southern Italians. This inversion of haplogroup dominance is incompatible with a Levantine paternal majority model for Ashkenazim and instead supports a Mediterranean paternal structure shaped by founder effects and long-term endogamy.

Demonstration of Ashkenazi Divergence from Levantine Paternal Structure

If Ashkenazi paternal ancestry were predominantly Levantine, Ashkenazi Y-chromosomal structure would be expected to resemble Levantine populations when evaluated using the same haplogroups and subclades. Levantine populations are commonly characterized by high frequencies of J1-M267, reduced representation of J2-M172 relative to J1, and comparatively low frequencies of E-M123. This pattern reflects demographic expansions that differ markedly from Southern European and Mediterranean Y-chromosomal structure.

When Ashkenazi Jews are compared against Levantine populations using these same markers, a clear divergence is observed. Ashkenazim show E-M123 and J2 at levels comparable to Southern Italian and Aegean populations, while J1 remains secondary. In contrast, Levantine populations show a reversal of this structure, with J1 dominating the paternal pool and E-M123 occurring at lower frequencies.

This contrast is summarized below.

Table 3. Y-DNA haplogroup structure in Ashkenazi Jews compared with Levantine populations (E, J1, J2 only).

Population	E-M78	E-M123	J1-M267	J2-M172
Ashkenazi Jews	5.2	11.7	14.6	23.2
Lebanese	~2 to 5	~4 to 5	~25 to 35	~10 to 15
Palestinians	~1 to 3	~2 to 4	~35 to 45	~10 to 15
Iraqis	~1 to 3	~2 to 3	~40 to 50	~10 to 15

Note. Values for Levantine and Iraqi populations are approximate ranges synthesized from Hammer et al. (2009), Behar et al. (2010), Nebel et al. (2001), and relevant studies cited in the text. All haplogroup frequencies are percent.

In this comparison, the Ashkenazi frequency profile differs from the J1-dominant pattern reported for several Levantine populations and overlaps Southern Italian ranges for J2 and E-M123. That relative ordering weighs against treating the sampled Ashkenazi paternal pool as a simple copy of a Levantine-majority profile. It does not, by itself, identify a unique geographic source.

Importantly, moderate elevation of J1 in Ashkenazim relative to some Southern Italian regions does not imply Levantine origin. Southern Italy itself exhibits substantial J1 heterogeneity, and Ashkenazi populations experienced strong founder effects and long-term endogamy that can amplify specific lineages. Drift can elevate J1 frequencies without implying descent from a population whose overall structure is J1-dominant. By contrast, Levantine J1 dominance reflects sustained demographic expansion that is not mirrored by the Ashkenazi paternal profile.

Direct comparison with both Southern European and Levantine populations therefore supports a Mediterranean rather than exclusively Levantine interpretation of these Y-chromosomal frequencies. Claims of precise geographic origin require denser regional sampling and higher-resolution subclades; the current evidence is best used as a consistency check on the autosomal comparator analysis.

Fine-Scale Southern European J1 Variation and Upper-Range Overlap

Fine-scale surveys of Southern European Y-chromosomal variation show that the range of haplogroup J1 observed in Southern Italy by Semino et al. (2004) does not represent an upper bound. When Southern Europe is sampled at micro-regional resolution (Di Giacomo et al., 2003;

Boattini et al., 2013), J1-M267 frequencies are highly heterogeneous and locally elevated across the broader Greco-Roman Mediterranean continuum. Province-level Italian datasets document multiple Southern Italian localities with J1 frequencies exceeding those reported in Semino et al., particularly in Sicily and parts of Campania. A later synthesis of Sicilian and Southern Italian genetic data (Sarno et al., 2017), incorporating these underlying regional surveys, confirms the presence of elevated local Southern Italian values while showing that these populations remain J2-dominant overall. These upper-range J1 values approach or overlap those observed in Ashkenazi Jews, demonstrating that moderate J1 enrichment in Ashkenazim does not exceed plausible Southern European source variation once regional structure and founder effects are taken into account.

Comparable J1 levels are also observed in parts of Central Italy, including Tuscany, indicating that elevated J1 is not restricted to the southern peninsula. Elevated values are likewise reported in the Aegean. Fine-scale sampling on Crete reports J1-M267 frequencies overlapping Southern Italian and Ashkenazi levels, including a value of approximately 10.5 percent at Nea Nikomedeia (King et al., 2008).

Table 4. Upper-range regional J1-M267 frequencies in Southern Italy and the Aegean.

Region	Population	J1-M267 (%)
Italy	Sicily, Agrigento	11.1
Italy	Western Sicily	10.13
Greece	Crete, Nea Nikomedeia	10.5
Italy	Trapani, Sicily	8.82
Italy	Campania, Benevento	8.3
Greece	Crete	8.3
Italy	Sicily, Southwest	7.0
Italy	Calabria, Tyrrhenian Calabria	6.8

Frequencies shown represent high reported regional J1-M267 values in Southern Italy and the Aegean. Agrigento and other synthesized values are reported in Sarno et al. (2017) and derive from underlying province-level datasets, including Boattini et al. (2013). Western Sicily and Trapani values are taken directly from Boattini et al. (2013). The Nea Nikomedeia value is reported in King et al. (2008). All listed populations retain a J2-dominant Mediterranean paternal structure.

Cohen Modal Haplotype Arguments and Comparator Omission

The Cohen Modal Haplotype literature, particularly Hammer et al. (2009), addresses the internal history of priestly lineages rather than the genome-wide origin of Ashkenazi Jews. Cohanim are genetically heterogeneous, J-P58 accounts for fewer than half of the sampled Cohanim lineages, and the original low-resolution CMH is not population-specific. These findings may inform the history of particular paternal lines but cannot establish a Levantine majority for the population as a whole.

Hammer et al. do not test J-P58 or CMH frequencies against Southern Italian, Sicilian, or Aegean populations. Consequently, the study cannot determine whether the observed lineages are Levantine-specific or part of a broader eastern and central Mediterranean distribution. Traditional

descent claims also lack an ancient genetic reference population against which geographic specificity could be tested.

This limitation is not unique to Hammer et al. Subsequent large studies focused on J-P58 phylogeography, including Chiaroni et al. (2010) and Fadhloui-Zid et al. (2011), restrict sampling largely to the Levant, Arabia, and North Africa and do not evaluate European or Mediterranean populations. As a result, these studies cannot distinguish Levantine specificity from broader Mediterranean continuity. When Southern Europe is properly sampled, J1 frequencies observed in Ashkenazi Jews fall within the upper range of documented Southern Italian and Aegean variation while remaining far below the J1-dominant structure characteristic of Levantine populations. Accordingly, J1 cannot be treated as evidence of a Levantine paternal majority in Ashkenazi Jews and is more parsimoniously interpreted as a minor lineage shaped by Mediterranean ancestry and subsequent founder effects.

Part IV. Interpretive Framework and Constraints on Inference

Genome-wide analyses provide the primary context for evaluating how Ashkenazi placement changes with reference design. When Southern European and central Mediterranean populations are included explicitly, several published PCA, clustering, and identity-by-descent analyses place Ashkenazi Jews near Southern Italian, Sicilian, Maltese, and Greek populations. Agreement among methods with different mathematical properties is informative, although proximity in a plot or distance matrix does not by itself identify a literal ancestral population.

Drineas et al. (2019) provide an illustrative Mediterranean-inclusive comparison. In their PCA analyses of Crete and European populations, Cretans overlap most closely with Peloponnesians, Sicilians, and Ashkenazi Jews. In that reference space, Ashkenazi Jews occupy a broader Southern European and central Mediterranean cluster rather than a simple Europe–Levant line. This example demonstrates the interpretive importance of including southern comparators; it is not, on its own, an admixture model.

Comparable sensitivity to reference composition is observed in large-scale ancient DNA studies that include modern comparative PCA. In Schuenemann et al. (2017), which incorporates extensive present-day reference populations alongside ancient Egyptian genomes, Ashkenazi Jews project within Southern European PCA space together with Southern Italians, Sicilians, and Maltese. These projections demonstrate that Mediterranean-inclusive reference sets allow evaluation of Ashkenazi placement relative to Southern Europe rather than forcing comparison between truncated European and Levantine poles.

Identity-by-descent analyses impose additional interpretive constraints because they quantify shared ancestry through inherited chromosomal segments rather than through geometric projection. Xue et al. (2017) report IBD sharing between Ashkenazi Jews and regional populations stratified by segment length, allowing separation of deeper shared ancestry from more recent shared ancestry. Within this framework, long shared segments reflect relatively recent common ancestry, whereas shorter segments capture older population structure. The relative presence or absence of long IBD segments therefore constrains models that posit substantial recent shared ancestry with specific source populations.

Taken together, these examples define interpretive boundaries rather than final conclusions. Apparent Ashkenazi “intermediacy” is shown to be contingent on reference population choice, comparator inclusion, and the mathematical properties of the analytical method employed. Within this framework, intermediate placement is treated as a hypothesis generated under specific modeling assumptions that must be tested explicitly under alternative, historically grounded reference configurations.

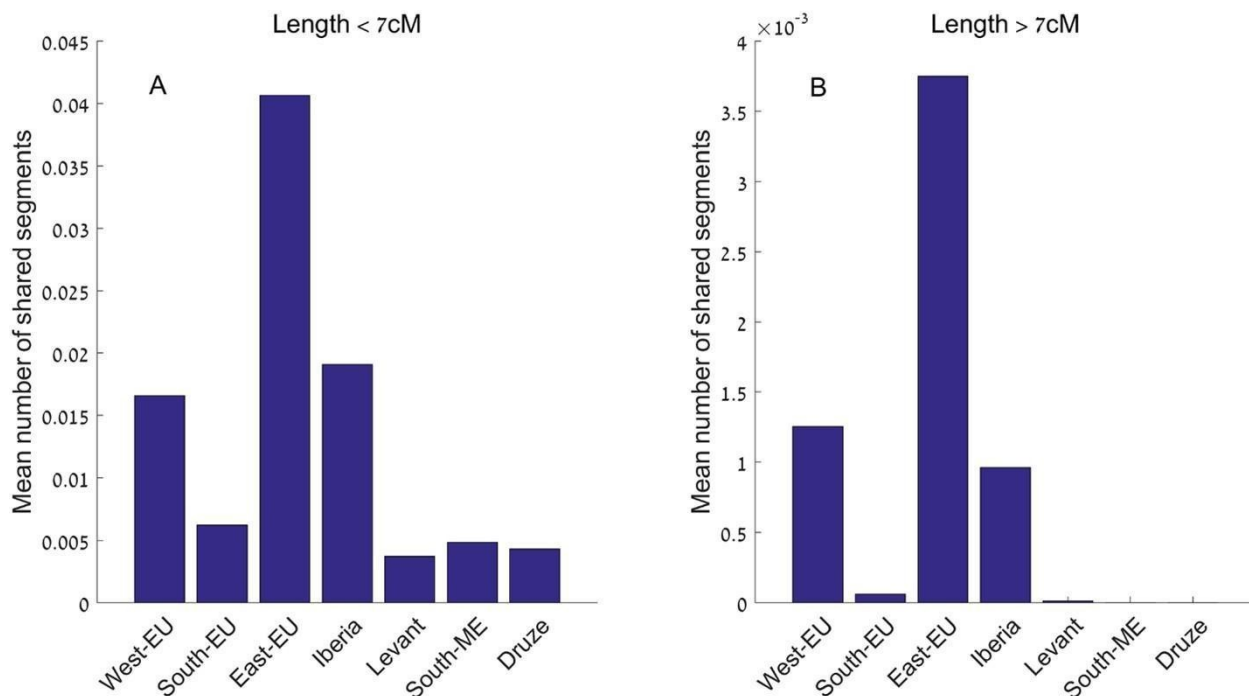


Figure 2. Identity-by-descent sharing between Ashkenazi Jews and regional populations from Xue et al. (2017). Panel A shows mean shared IBD segments shorter than 7 cM, reflecting deeper shared ancestry, while Panel B shows mean shared IBD segments longer than 7 cM, reflecting more recent shared ancestry. The distribution of shared segment lengths illustrates how IBD-based analyses constrain inferences about recent versus older shared ancestry across regional populations and provide an interpretive benchmark for evaluating models that posit substantial recent Levantine versus European contributions.

Part V. Convergent Autosomal Modeling Across Independent Analytical Frameworks

Several autosomal studies using different datasets and methods report a large Italian or central Mediterranean-associated component in Ashkenazi Jews. The estimates are not statistically independent in every respect, and their population labels depend on available proxies. Their value is therefore in the recurring regional ordering, not in treating any one point estimate as definitive.

Lerga-Jaso et al. (2025) applied a high-resolution local ancestry inference framework (Orchestra) to Ashkenazi Jewish genomes. Orchestra combines a recombination-based first-pass layer grounded in identity-by-descent assumptions with a deep-learning smoothing module integrating convolutional and attention-based architectures. The method assigns ancestry at fine genomic scale by leveraging the expectation that individuals share longer haplotypes with more closely related populations and shorter haplotypes with more distant ones. Using this framework, the authors inferred Ashkenazi ancestry as predominantly Italian, estimating ~68% Italian ancestry, followed by Levantine (16.6%), Iraqi-Iranian-Caucasian-Turkish (7.2%), Greek-Balkan (2.4%), and Eastern European (1.7%) components.

A complementary estimate is provided by Waldman et al. (2022), who used qpAdm to evaluate historically plausible admixture models for Ashkenazi Jews. Under their preferred model, Ashkenazi ancestry is inferred as ~65% Southern Italian, ~19% Lebanese, and ~16% Eastern European. I do not interpret the Lebanese component as a stand-alone Levantine origin term. Lebanese populations carry substantial eastern Mediterranean structure and are shifted toward Southern European genetic space relative to Arabian-associated Levantine poles. Accordingly, a Lebanese proxy in qpAdm is compatible with a broader Mediterranean continuum and should not be treated as evidence for a Levantine-majority model.

Brace et al. (2022) analyzed genomes from a medieval Jewish burial assemblage in Norwich, England. The individuals show Ashkenazi-associated hereditary variation before the twelfth century, and one reported proxy model approximated the medieval group with a majority Sicilian contribution and a secondary Turkish Jewish contribution. Because the sample is small and proxy labels are model-dependent, this result is best read as evidence that a substantial central or eastern Mediterranean-associated component was already present in a medieval Jewish population, not as a precise ancestry census.

Across local-ancestry inference, qpAdm modeling, and medieval ancient-DNA analysis, the broad ordering is similar: Italian or closely related central Mediterranean proxies account for a large

component, while eastern Mediterranean and later Eastern European proxies account for smaller components. The exact values vary because the methods partition shared Mediterranean structure differently. This convergence motivates, but does not replace, the controlled reference-panel tests in Part IX.

Table 5. Convergent autosomal estimates of Southern Italian and central Mediterranean ancestry in Ashkenazi Jews.

Study	Methodology	Data type	Southern Italian / Mediterranean component	Additional components (Local ancestry)
Lerga-Jaso et al. (2025)	IBD-based inference (Orchestra; plus deep-learning smoothing)	Modern genomes	~68% Italian	~16.6% Levantine; ~7.2% Iraqi-Iranian-Caucasian-Turkish; ~2.4% Greek-Balkan; ~1.7% Eastern European
Waldman et al. (2022)	qpAdm (historically constrained model)	Modern genomes	~65% Southern Italian	~19% Lebanese; ~16% Eastern European
Brace et al. (2022)	qpAdm anchored to Medieval Jewish; medieval ancient DNA (Chapelfield) plus modern	Medieval ancient DNA plus modern	~67% Sicilian (Southern Italian proxy)	~33% Turkish Jewish; ~0% French; ~0% Polish

Table 5 note. Components are reported as proportions of modeled ancestry (percent). Point estimates differ because methods partition ancestry differently and qpAdm results depend on the proxy populations available. In particular, Lebanese populations are eastern Mediterranean and shifted toward Southern European genetic space relative to Arabian-associated Levantine poles; therefore, a Lebanese component in qpAdm should not be equated with a Levantine-majority origin claim. Across all approaches, Southern Italy remains the dominant autosomal source, with additional components reflecting broader eastern Mediterranean continuity and later Eastern European admixture.

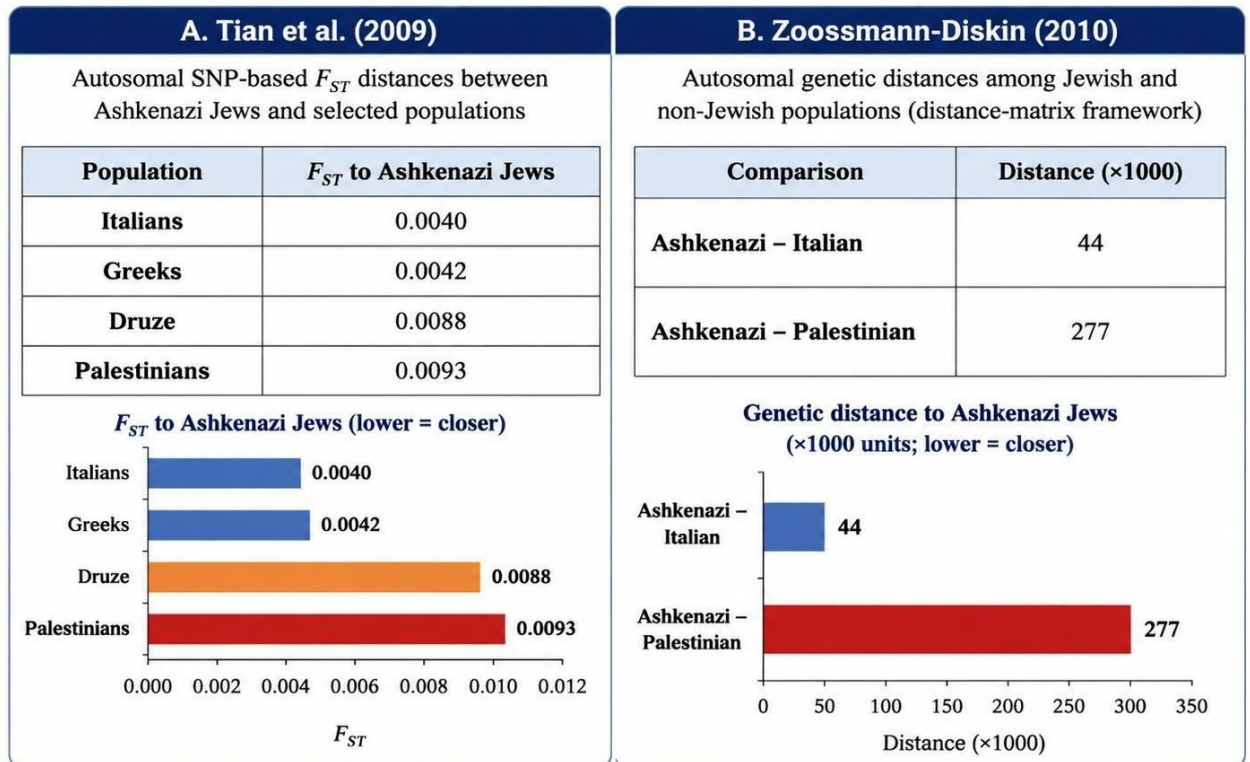


Figure X. Independent autosomal F_{ST} analyses converge on the same ordering of affinities.

Tian et al. (2009) found the smallest autosomal distances between Ashkenazi Jews and Italians (0.0040) and Greeks (0.0042), with substantially larger distances to Druze (0.0088) and Palestinians (0.0093). Zoosmann-Diskin (2010) similarly found Ashkenazi Jews much closer to Italians (44) than Palestinians (277). Despite differing datasets, population panels, and analytical frameworks, both studies recover the same pattern: Ashkenazi Jews are genetically closest to **Southern Europeans and substantially more distant from Levantine populations.**

Table 6. Autosomal F_{ST} proximity of Ashkenazi Jews in two independent studies.

A. Tian et al. (2009): SNP-based autosomal F_{ST}		B. Zoossman-Diskin (2010): autosomal genetic distance matrix ($\times 1000$)	
Comparison population	F_{ST}	Comparison population	Distance ($\times 1000$)
Italians	0.0040	Italians	44
Greeks	0.0042	Greeks	105
Spanish	0.0056	Turks	170
Tuscans	0.0066	Germans	131
Germans	0.0072	French	144
Druze	0.0088	British	238
Palestinians	0.0093	Poles	195
Irish	0.0109	Russians	230
Swedes	0.0120	Palestinians	277
Russians	0.0137		
Basque	0.0144		

Table 6 note. Use NR for values not reported. Use ~ for approximate values. Both studies independently identify Italians and Greeks as the closest populations to Ashkenazi Jews, with Palestinians substantially more distant. Ashkenazi–Southern European distances fall in the ~0.004 to ~0.005 F_{ST} range typical of closely related regional populations, whereas Ashkenazi–Palestinian distances approach or exceed ~0.009, indicating markedly deeper divergence under these panels. These results contradict claims that Ashkenazi Jews occupy a genetic midpoint between Europe and the Levant; apparent intermediacy arises when Southern Europe is excluded or replaced with genetically inappropriate proxies.

Part VI. Integrated Interpretation and Implications for Models of Ashkenazi Origins

Across the analyses reviewed or generated here, Ashkenazi Jews repeatedly show close autosomal affinity to Southern European and central Mediterranean populations. Pairwise F_{ST} comparisons often rank Southern Italian and Greek populations among the closest sampled groups, whereas Levantine comparisons are more distant in the reported panels. This ordering is compatible with substantial central Mediterranean ancestry, but genetic distance alone cannot distinguish shared ancestry from all alternative demographic histories.

qpAdm models reviewed or generated here likewise assign a large component to Southern Italian or related central Mediterranean proxies, with additional eastern Mediterranean and Eastern European components. Proxy choice matters: omitting Southern Europe or substituting north-shifted Italian and drifted island populations can divide the missing Mediterranean direction between the available European and Levantine sources. The matched Global25 omissions test that mechanism directly and show that the apparent cline strengthens as the central Mediterranean source space is removed.

A limited consistency check is provided by the CCR5-Δ32 allele, which follows a pronounced European north–south frequency cline and is rare or absent in several sampled Levantine and Arabian populations (Stephens et al., 1998). Reported Ashkenazi frequencies fall within the European distribution. This pattern is compatible with European acquisition followed by amplification during founder events and endogamy, but one locus cannot estimate genome-wide ancestry.

The CCR5-Δ32 distribution is therefore inconsistent with treating all Ashkenazi ancestry as a direct sample of populations in which the allele was absent, but it does not exclude substantial eastern Mediterranean or Levantine-related ancestry. Haplotype and coalescent analyses place the allele's rise within-European demographic history (Stephens et al., 1998). I use it only as an ancillary check on the broader autosomal model, not as an independent falsification of a population-level ancestry proportion.

More generally, Ashkenazi disease-associated founder variants demonstrate how bottlenecks and endogamy can amplify particular alleles. Such variants are not geographically diagnostic without direct comparative and haplotypic evidence, and they are not used here to estimate ancestry. Their relevance is methodological: single-locus frequencies can diverge sharply from genome-wide proportions under strong founder effects.

Taken together, these lines of evidence support a bounded conclusion. Apparent Ashkenazi placement on a Europe–Levant cline is not invariant; it changes when Southern European and central Mediterranean comparators are included or removed. The most direct evidence is the matched panel response, while the published PCA, FST, qpAdm, identity-by-descent, and ancient-DNA results provide external context. This pattern supports a large Italian Jewish and central Mediterranean-associated component, additional eastern Mediterranean and Levantine continuity, and later Northern and Eastern European contributions.

Table 7. CCR5-Δ32 allele frequencies in Ashkenazi Jews and comparative populations.

Population	Sample size	CCR5-Δ32 allele frequency
Swedish	131	0.137
Estonian	158	0.133
Polish	30	0.133
French	230	0.089
Italian	172	0.055
Greek	160	0.044
Ashkenazi Jews	503	0.097
Lebanese	51	0.000
Saudi Arabian	100	0.000
Chinese	40	0.000
African populations	various	0.000

Table 7 note. Allele frequencies are reported as proportions (0 to 1) as presented in Stephens et al. (1998). CCR5Δ32 frequencies follow a European north-south cline. Ashkenazi Jews fall within the European distribution and contrast with Levantine and Arabian comparators reported as 0.000 in this dataset.

Part VII. Historical and Geographic Context of Jewish Settlement in Southern Italy and the Central Mediterranean

Genetic models that use Northern Italian or Tuscan populations as generic Italian proxies should also be evaluated against the historical geography of Jewish settlement. Jewish communities were established across Southern Italy, Sicily, and the wider central and eastern Mediterranean before the formation of medieval Ashkenazi communities in northern Europe. These are also the regions that

recur in the Mediterranean-inclusive PCA and FST comparisons and in the matched Global25 tests presented in Part IX (Feldman, 1993; Gruen, 2002; Rutgers, 1995).

Jewish presence in Southern Italy and Sicily is documented from antiquity. Archaeological, epigraphic, and literary evidence attests to communities on the southern mainland and in Sicily during the Roman and Late Antique periods (Rutgers, 1995). In Sicily, settlement is documented in Syracuse, Palermo, Catania, Agrigento, and Mazara del Vallo, with continuity through Byzantine and medieval rule until the expulsions of the late fifteenth century (Roth, 1946; Rutgers, 1995). This record establishes the central Mediterranean as a historically plausible context for later Jewish population formation; it does not, by itself, quantify genetic contribution.

On the southern Italian mainland, Jewish communities are documented from antiquity in Apulia, Calabria, Campania, and Lucania. Roman-period and Late Antique evidence supports Jewish settlement in Bari, Brindisi, Oria, Otranto, Trani, Reggio Calabria, Catanzaro, Squillace, Naples, Pozzuoli, and Capua (Roth, 1946; Rutgers, 1995). Oria emerges as a major Jewish intellectual center in Late Antiquity and the early medieval period, underscoring the depth and continuity of Jewish life in southern Italy well before Ashkenazi ethnogenesis (Roth, 1946; Bonfil, 1994). These communities participated in a Greek- and Latin-speaking Mediterranean network with sustained ties to southern Greece, the Aegean islands, and Asia Minor, where Jewish settlement is likewise securely attested from the Hellenistic period onward (Feldman, 1993; Gruen, 2002).

Large documented Jewish centers in Northern Italy and Tuscany expanded later than several southern and Sicilian communities and were connected to movements within the Italian peninsula and the wider Mediterranean (Roth, 1946; Bonfil, 1994). Northern Italian and Tuscan populations may therefore be useful modern comparators, but they should not be treated as the only or automatically preferred proxies for an ancestral Italian component. Historically relevant southern comparators must be tested alongside them.

Distinctive Ashkenazi communities emerge in northern France and the Rhineland during the early medieval period. Historical scholarship identifies Italy and the central Mediterranean as plausible staging areas for northward movement along established communication and trade routes (Chazan, 1987; Grossman, 1995). The later Ashkenazi bottleneck and long-term endogamy could then reshape frequencies without erasing the older Mediterranean-associated autosomal signal. History provides plausibility and directionality here, not a numerical ancestry estimate.

This historical geography aligns with genome-wide genetic results. Autosomal PCA and FST analyses repeatedly place Ashkenazi Jews closest to southern Italian, Sicilian, and related central Mediterranean populations, with substantially greater distances to Levantine populations (Tian et al., 2009; Zoosmann-Diskin, 2010). Italian Jewish populations provide an important internal control. These communities descend from the southern Italian and Sicilian Jewish populations documented from antiquity and, in general, do not exhibit the same severe demographic bottleneck as Ashkenazi Jews. In autosomal analyses, Italian Jews cluster within southern Italian and central Mediterranean genetic space and overlap Ashkenazi Mediterranean placement once drift is accounted for, supporting continuity rather than convergence through recent admixture (Atzmon et al., 2010; Zoosmann-Diskin, 2010).

Accordingly, Northern Italian or Tuscan proxies should not stand alone as representations of Italian ancestry in Ashkenazi models. Experiments 10–12 quantify the consequence. With Italki and central Mediterranean sources present, the strict block totals 54.6. Removing Italki worsens fit and transfers most of its coefficient to related Mediterranean sources. Removing the wider central Mediterranean source space worsens fit by 47.18% while Lombardy rises from 9.8 to 38.0 and Levantine continuity rises from 22.8 to 52.0. The result is the manuscript's central methodological finding: within the matched panel, omission of the relevant central Mediterranean comparators creates an apparently stronger Northern Italian–Levantine mixture even though that two-pole solution fits the target substantially worse.

Part VIII. Independent qpAdm and FST Analyses with External Validation

I conducted the qpAdm and autosomal FST analyses reported here through the IllustrativeDNA Admix Lab web interface, as specified in Part I. These results provide an author-generated comparison with the published PCA, FST, and identity-by-descent literature. They are not presented as independent command-line ADMIXTOOLS replication. The separate Global25 suite in Part IX addresses reference-panel sensitivity through matched omissions, reciprocal models, regional replication, and ancient positive controls.

Table 9. qpAdm results for Jew_Ashkenazi.HO (author-generated).

Model	Italian (%)	Leb. (%)	Rus. (%)	P	Fit
Italian_South.HO + Lebanese.HO + Russian.DG	69.97	22.06	7.97	0.1445	Yes
Italian_North.HO + Lebanese.HO + Russian.DG	54.53	42.57	2.90	0.0256	No
Italian_Central.HO + Lebanese.HO + Russian.DG	61.33	33.83	4.84	0.0514	Yes

Note. Under the interface's displayed threshold, the Southern Italian model is feasible ($p = 0.1445$), the Northern Italian model is infeasible ($p = 0.0256$), and the Central Italian model is marginally feasible ($p = 0.0514$). Coefficients are conditional on this source and right-population specification and should be compared with the full sensitivity analysis rather than read as unique historical proportions.

Table 10. Autosomal FST distances to Jew_Ashkenazi.HO (author-generated).

Population	FST distance	Standard error
Cretan.DG	0.0049	0.00124
Italian_South.HO	0.0049	0.00033
Sicilian.HO	0.0052	0.00041
Greek.HO	0.0055	0.00035
Maltese.HO	0.0066	0.00045
Jew_Moroccan.HO	0.0091	0.00058
Jew_Iraqi.DG	0.0122	0.00138
Jew_Iranian.HO	0.0125	0.00050
Jew_Tunisian.HO	0.0127	0.00056
Jew_Libyan.HO	0.0132	0.00053
Jew_Georgian.HO	0.0133	0.00056
Jew_Yemenite.HO	0.0174	0.00054

Note. Cretan and Southern Italian comparisons share the lowest displayed distance, followed by Sicilian, Greek, and Maltese populations. The sampled non-European Jewish comparators are more distant in this panel. FST ranks relative affinity; it does not by itself identify a historical source or admixture proportion.

Table 11. F_{ST} calibration benchmarks.

Population pair	F_{ST} distance	Interpretation
Italian_South.HO vs Maltese.HO	0.0036	Closely related central Mediterranean populations
Jew_Ashkenazi.HO vs Italian_South.HO	0.0049	Strong regional affinity
IBS_CanaryIslands.DG vs Spanish.HO	0.0038	Iberian-derived population with additional North African ancestry

Genome-wide autosomal and ancient DNA studies show that modern Canary Islanders derive a strong majority of their ancestry from Iberian sources, with a minority Indigenous North African component often estimated at approximately 15 to 25 percent (Botigué et al., 2013; Rodríguez-Varela et al., 2017). Despite this admixture, Canary Islanders cluster tightly with Iberians in PCA space and show very low F_{ST} distances to Spanish populations. This provides an external calibration demonstrating that very low F_{ST} values on the order of 0.003 to 0.004 are compatible with strong shared ancestry in the presence of admixture and drift. Under this calibration, the Ashkenazi–Southern Italian F_{ST} distance of 0.0049 is consistent with strong regional affinity rather than weak relatedness or an intermediate position.

In addition, PCA frameworks discussed elsewhere in this study show that Ashkenazi Jews overlap extensively with Southern Italian populations in Mediterranean-inclusive space. In comparative terms, this overlap is not weaker than the Iberian signal observed for the Canary Islands, even though Canary Islanders retain a higher proportion of direct Iberian ancestry. This pattern is consistent with a Southern Italian and central Mediterranean anchoring of Ashkenazi autosomal variation, with drift and endogamy shaping within-cluster dispersion rather than shifting Ashkenazi Jews toward Levantine genetic space.

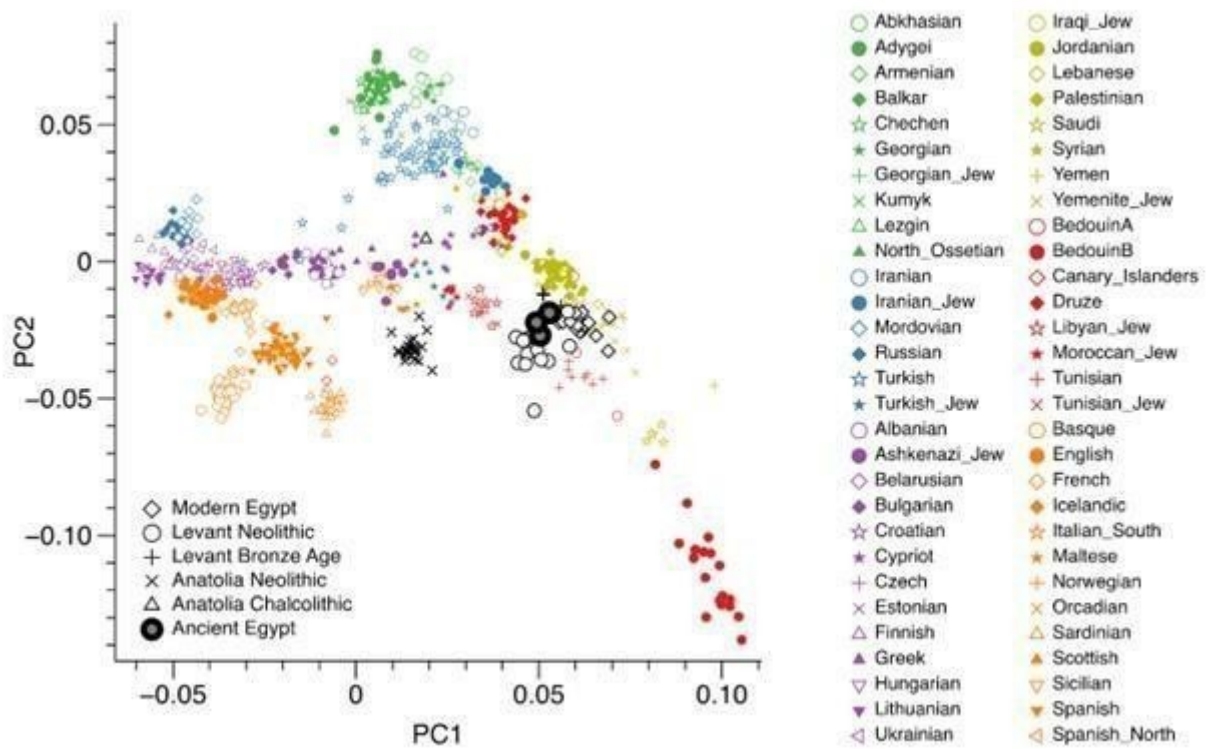


Figure 3. Principal component analysis reproduced from Schuenemann et al. (2017), shown for comparative context. Roman-period and earlier Egyptian individuals cluster with Mediterranean populations, while post-Roman Egyptian samples shift toward increased Sub-Saharan African ancestry. The figure illustrates that PCA space can capture post-antique admixture trajectories without redefining core Mediterranean structure. In comparable Mediterranean-inclusive PCA frameworks discussed elsewhere in this study, Ashkenazi Jews align with Southern European and central Mediterranean populations rather than along a Levantine axis, consistent with the Southern Italian proximity signal observed in qpAdm and FST.

Summary of Author-Generated qpAdm and FST Validation Analyses

Across the reported qpAdm models, autosomal FST distances, and Mediterranean-inclusive PCA comparisons, Southern Italian, Sicilian, Cretan, and Greek populations repeatedly rank among the closest comparators to Ashkenazi Jews. Northern Italian proxies produced poorer fit or were infeasible in the tested qpAdm specifications. These results support a central Mediterranean-associated model within the tested datasets; they do not establish that one present-day comparator is the literal ancestral population.

These independently generated qpAdm and FST results support a central Mediterranean anchoring model but do not by themselves identify how alternative reference panels redistribute the same signal. Part IX addresses that question explicitly with fixed panels, omission tests, reciprocal Italki models, regional replications, and ancient Levant controls.

Part IX. Global25 Reference-Panel Sensitivity, Omission Tests, Reciprocal Controls, and Ancient Levant Validation

I evaluated reference-panel sensitivity with a linked suite of fifteen Global25 experiments. The suite is not treated as fifteen independent ancestry estimates. It combines fixed-source prehistoric controls, broad modern competition panels, matched source-omission tests, direct-distance diagnostics, reciprocal Italki models, regional Ashkenazi replications, and ancient Levant positive controls. Interpretation rests on repeated behavior under matched source-panel changes rather than on one coefficient, coordinate, or plot.

Analytical Design and Limits

All mixture experiments use declared scaled Global25 population averages and nonnegative constrained modeling. The calculator was run with cycles at 0.25X, distance column off, printed zeroes off, fast mode off, and aggregate mode on. Experiments 1 and 8 use prehistoric components in fixed panels; prehistoric references are not mixed into the modern population competitions. Experiments 9 and 15 serve as method controls by asking whether ancient Levantine targets are assigned primarily to appropriate Levantine comparators. Exact source labels, targets, raw outputs, grouped summaries, screenshots, and checksums are preserved in the accompanying repository.

Displayed coefficients are constrained similarity weights conditional on the competing sources. They are not literal historical ancestry percentages, and no formal confidence intervals or admixture dates are inferred. Correlated populations can divide or replace the same ancestry direction; therefore, a zero coefficient means that another source represented the target more efficiently in that panel, not that historical ancestry from the zeroed population was absent. The strongest inferences come from fit changes, category-level redistribution, reciprocal results, regional persistence, and positive-control performance. Small averages, including Italki $n=9$ and several regional or ancient targets with $n=2$ to $n=4$, require caution.

Prehistoric-Component Controls: Experiments 1 and 8

Experiment 1 applies the same seven prehistoric sources to 196 declared Italki, regional Ashkenazi, Southern Italian, Maltese, Cretan, and Dodecanese targets, with a separate matched pooled Ashkenazi run. The forty Southern Italian individuals average 7.57 percent displayed Israel_Epipaleolithic_Natufian_ ($n=2$) and span 2.4–12.2. Pooled Ashkenazi Jews score 12.8 and the ten regional Ashkenazi averages span 11.4–15.0; Italki scores 16.2. The 9.8-point internal Southern Italian range is 2.45 times the 4.0-point gap between the Southern Italian maximum and Italki. Malta spans 4.0–16.4 and Crete 0.0–13.4. A restricted regional mean therefore cannot serve as a biological maximum, and one prehistoric component cannot decide historical origin.

Experiment 8 places ancient and modern Levantine controls, Italki, pooled Ashkenazi, Southern Italian, Maltese, and Aegean averages under a shared prehistoric panel. Russia_Karelia_Mesolithic_(EHG)_ ($n=15$) is exactly 0.0 in every ancient Levantine row and is 0.0 in Samaritan, Palestinian Arab Christian, and Lebanese Arab Christian; Druze Israel is the sole modern Levantine exception at 1.6. The same displayed coefficient is 8.6 in Italki and 14.8 in Ashkenazi Jews, with central Mediterranean controls ranging from 5.8 to 14.8. These prehistoric labels do not imply recent ethnic Russian ancestry; they identify a coordinate contrast within the declared panel.

Two additional components produce the same separation. Iran_N_Ganj_Dareh_ ($n=7$) is 11.4 in Italki and 9.8 in Ashkenazi Jews, compared with approximately 16–25 in most later ancient and modern Levantine rows; Israel_CA_Peki'in_ ($n=20$) is an acknowledged early exception at 8.0. Turkey_Anatolia_N_Ceramic_Barcin_ ($n=22$) is 53.2 in Italki and 51.2 in Ashkenazi Jews, compared with 28.6–47.2 in the ancient Levant and 39.2–44.0 in the modern Levantine controls. Natufian is intermediate: 16.2 in Italki and 12.8 in Ashkenazi Jews, above most central Mediterranean controls but below most Levantine rows. The multicomponent pattern supports retained Levant-related continuity inside a predominantly central Mediterranean structure rather than direct continuity from the tested ancient Levantine profiles.

Table 12. Experiment 8 component contrast under the fixed prehistoric source panel. Values are displayed coefficients or observed ranges among the listed population averages.

Comparison	EHG	Iran_N	Anatolia_N	Natufian
Italki_Jew_($n=9$)	8.6	11.4	53.2	16.2
Ashkenazi_Jew_($n=1209$)	14.8	9.8	51.2	12.8
Ancient Levant targets	0.0 in all	8.0–25.2	28.6–47.2	25.0–48.6
Modern Levant controls	0.0; Druze 1.6	21.0–21.4	39.2–44.0	20.8–30.8
Southern Italy, Malta, Aegean	5.8–14.8	7.0–13.4	56.0–60.2	5.0–9.8

Table note. EHG denotes Russia_Karelia_Mesolithic_(EHG)_ ($n=15$); Iran_N denotes Iran_N_Ganj_Dareh_ ($n=7$); Anatolia_N denotes Turkey_Anatolia_N_Ceramic_Barcin_ ($n=22$); and Natufian denotes Israel_Epipaleolithic_Natufian_ ($n=2$). The modern Levantine EHG entry is 0.0 except for Druze Israel at 1.6.

Full Modern Competition: Experiments 2 and 10

Experiment 2 provides the first broad modern competition. The pooled Ashkenazi target has distance 0.00948576 and retains 58.2 Italki, 18.6 Rhodian Greek, 5.8 Maltese, 3.0 central Sicilian, 0.4 Cretan, 9.4 Belarusian, and 4.6 Swedish. Grouped descriptively, 86.0 is assigned to Italian Jewish, central Mediterranean, and Aegean sources and 14.0 to Northern and Eastern Europe. The result is not a final ancestry estimate; it establishes that Italki is the largest individual source when relevant Mediterranean and northern sources compete.

Experiment 10 expands the panel by adding Samaritan, Druze Israel, Palestinian Arab Christian, and Lebanese Arab Christian controls. At distance 0.00819655, Italki_Jew_(n=9) remains the largest individual source at 40.8. Italki, Maltese, and western Sicilian total 54.6; adding the separate 9.8 Lombardy coefficient produces a broader Italian and central Mediterranean total of 64.4. Druze Israel contributes 22.8, while German and Russian Smolensk total 12.8. Lombardy is not classified as Southern Italian. The key ordering is an Italian Jewish and central Mediterranean majority, a meaningful but secondary Levantine-continuity component, and a smaller Northern and Eastern European layer.

Omission and Proxy-Inflation Tests: Experiments 3, 5, 11, and 12

Experiments 3 and 5 establish the earlier omission behavior. Removing Northern and Eastern European sources in Experiment 3 raises distance from 0.00948576 to 0.01293736 and reallocates 99.6 of the displayed solution to Sicilian, Maltese, and Cretan references, with 0.4 Friulian. Removing Italki instead in Experiment 5 produces distance 0.01143262 and a 93.6 Maltese, Sicilian, and Aegean total versus 6.4 Belarusian. These panels show both that a minority Northern and Eastern European layer improves fit and that the larger Italki-associated signal is replaced chiefly within the Mediterranean comparison space.

Experiment 11 is the matched Italki omission test for Experiment 10. Removing only Italki worsens distance from 0.00819655 to 0.00922012, a relative deterioration of 12.49 percent. Maltese rises by 22.6 points, western Sicily by 7.2, and Lombardy by 4.6, for a 34.4-point Italian and central Mediterranean increase. The Levantine-continuity block rises by 9.8, while Northern and Eastern Europe falls by 3.4. These changes sum exactly to the omitted 40.8 Italki coefficient. Thus, the panel replaces Italki first through related Mediterranean and Italian sources, secondarily through Levantine controls, and not through Germany, Poland, or Russia.

Experiment 12 removes Italki together with Southern Italian, Sicilian, Maltese, and Aegean sources. Distance rises to 0.01206331, a 47.18% deterioration relative to Experiment 10 and 30.84% relative to Experiment 11. Lombardy rises from 9.8 to 38.0, the Levantine-continuity block rises from 22.8 to 52.0, and Northern and Eastern Europe remains 10.0. Relative to Experiment 10, the 28.2-point Lombardy increase plus the 29.2-point Levantine increase minus the 2.8-point Northern and Eastern European decrease equals the omitted 54.6-point Italki–Maltese–Sicilian block. The missing central Mediterranean direction is therefore partitioned between the two remaining proxy poles. Within this matched series, the stronger Northern Italian–Levantine cline is a measurable omission effect, not an improved representation of the target.

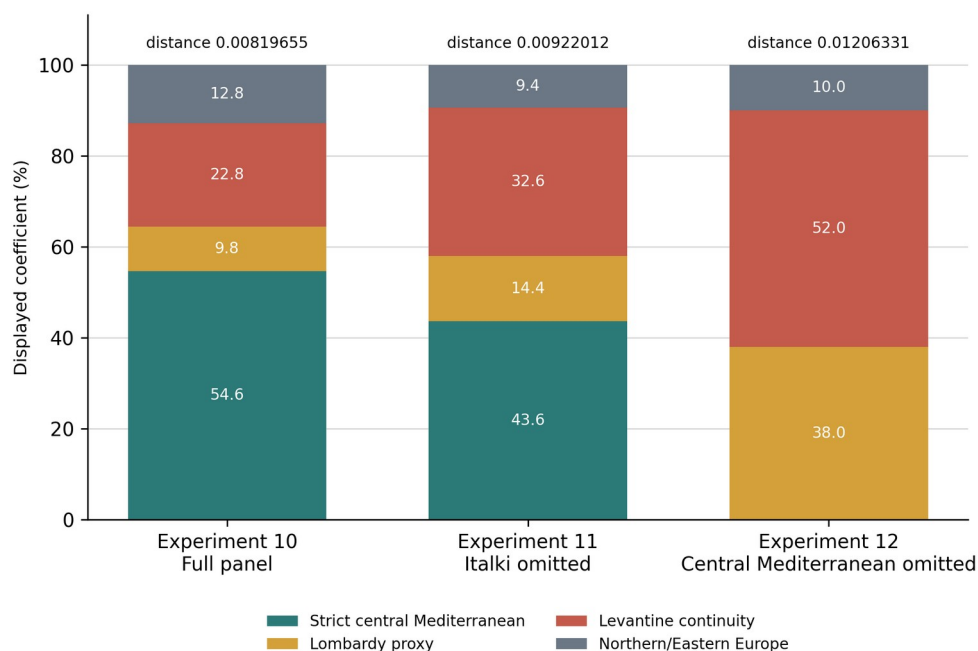


Figure 4. Matched source-panel comparison for Experiments 10–12. With the full panel, the strict central Mediterranean block is 54.6 and distance is 0.00819655. Removing Italki shifts the block to related sources and worsens fit. Removing the broader central Mediterranean source space eliminates that block, raises Lombardy to 38.0 and Levantine continuity to 52.0, and worsens distance to 0.01206331. Values are displayed coefficients, not literal ancestry percentages.

Direct-Distance and Reciprocal Italki Falsification: Experiments 4, 6, and 13

Experiment 4 is a direct-distance diagnostic rather than a mixture model. Maltese is nearest to the pooled Ashkenazi average at 0.01918750, followed by Lasithi Cretans, Mani Greeks, general Cretans, and Italki Jews. Tuscany begins the Northern Italian and Tuscan comparison set at 0.04672433, and the named Northern and Eastern European references are farther still. Direct distance cannot assign historical ancestry, but it shows that the populations chosen as central Mediterranean sources are substantially closer proxies than the Northern Italian alternatives.

Experiment 6 reverses the target and models Italki. At distance 0.01732664, the displayed solution is 52.6 Maltese, 44.6 Rhodian Greek, 2.4 eastern Sicilian, and 0.4 Reggio Calabria. No Northern Italian, Tuscan, German, Polish, Russian, or pooled Ashkenazi source is retained. This reciprocal result shows that the central Mediterranean placement exists independently in an Italian Jewish population and does not require later Ashkenazi residence north of the Alps.

Experiment 13 repeats the reciprocal design in the expanded modern panel, excludes Ashkenazi Jews as a source, and permits Levantine and Northern and Eastern European competitors. At distance 0.00984550, the Italki target retains 39.8 Maltese, 28.8 Calabria, 2.0 western Sicily, 4.2 Lombardy, 23.4 Samaritan, and 1.8 Palestinian Arab Christian. The strict central Mediterranean total is 70.6, the broader Italian and central Mediterranean total is 74.8, Levantine continuity is 25.2, and Northern and Eastern Europe is 0.0 in the displayed solution. The zero does not prove historical absence; the falsification is that northern sources were available but were unnecessary to reproduce the Italki profile.

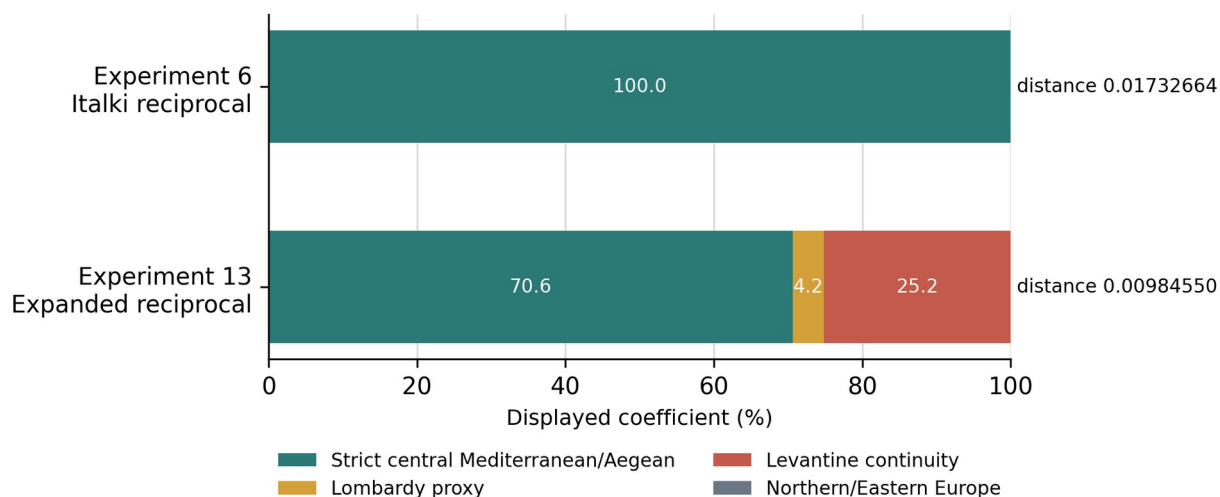


Figure 5. Reciprocal Italki models in Experiments 6 and 13. The original panel resolves entirely through central Mediterranean and Aegean sources. The expanded panel separates a 70.6 strict central Mediterranean block, a 4.2 Lombardy coefficient, and 25.2 Levantine continuity, with no Northern or Eastern European coefficient retained. Distances and displayed coefficients are panel-specific diagnostics.

Regional Ashkenazi Stability: Experiments 7 and 14

Experiment 7 applies one modern panel to ten named regional Ashkenazi population averages. Italki is the largest individual source in all ten, ranging from 41.2 in Germany to 70.2 in Austria, with a displayed average of 57.0. The grouped Italian Jewish, Sicilian, Maltese, and Aegean total averages 89.5, compared with 10.6 Northern and Eastern European. No Northern Italian or Tuscan source receives a displayed nonzero coefficient. Although Austria, Latvia, and Romania each have $n=2$, the same category-level structure appears in larger Polish, Ukrainian, Belarusian, Lithuanian, German, and Russian averages.

Experiment 14 repeats the ten-region test with the expanded Experiment 10 panel and modern Levantine controls. Italki is retained in all ten targets and is the largest individual source in nine; Lithuania is the exception, although its strict central Mediterranean block remains a majority. The strict block exceeds 50 in every region, spans 50.8–76.4, and averages 58.6. Adding Lombardy gives a broader Italian and central Mediterranean average of 62.5. Levantine continuity averages 23.5 and Northern and Eastern Europe 14.0. These values closely reproduce

The regional distributions also constrain a host-population explanation. Ashkenazi Poland retains 53.6 Italki and 0.0 Polish; Ashkenazi Russia retains 41.6 Italki and 9.0 Russian Smolensk; Ashkenazi Ukraine retains 45.2 Italki and 7.6 Russian Smolensk; and Ashkenazi Germany retains 31.8 Italki and 6.8 German. Exact coefficients fluctuate because correlated Mediterranean proxies substitute for one another. The category weights also differ in direction, not only magnitude. In the German target, the Levantine and Northern/Eastern European allocations form strongly opposing full-dimensional displacement vectors whose weighted effects are nearly equal. Their combined effect leaves the reconstructed target close to the central Mediterranean position. The displayed Levantine coefficient therefore cannot be interpreted directly as an additional historical ancestry percentage.

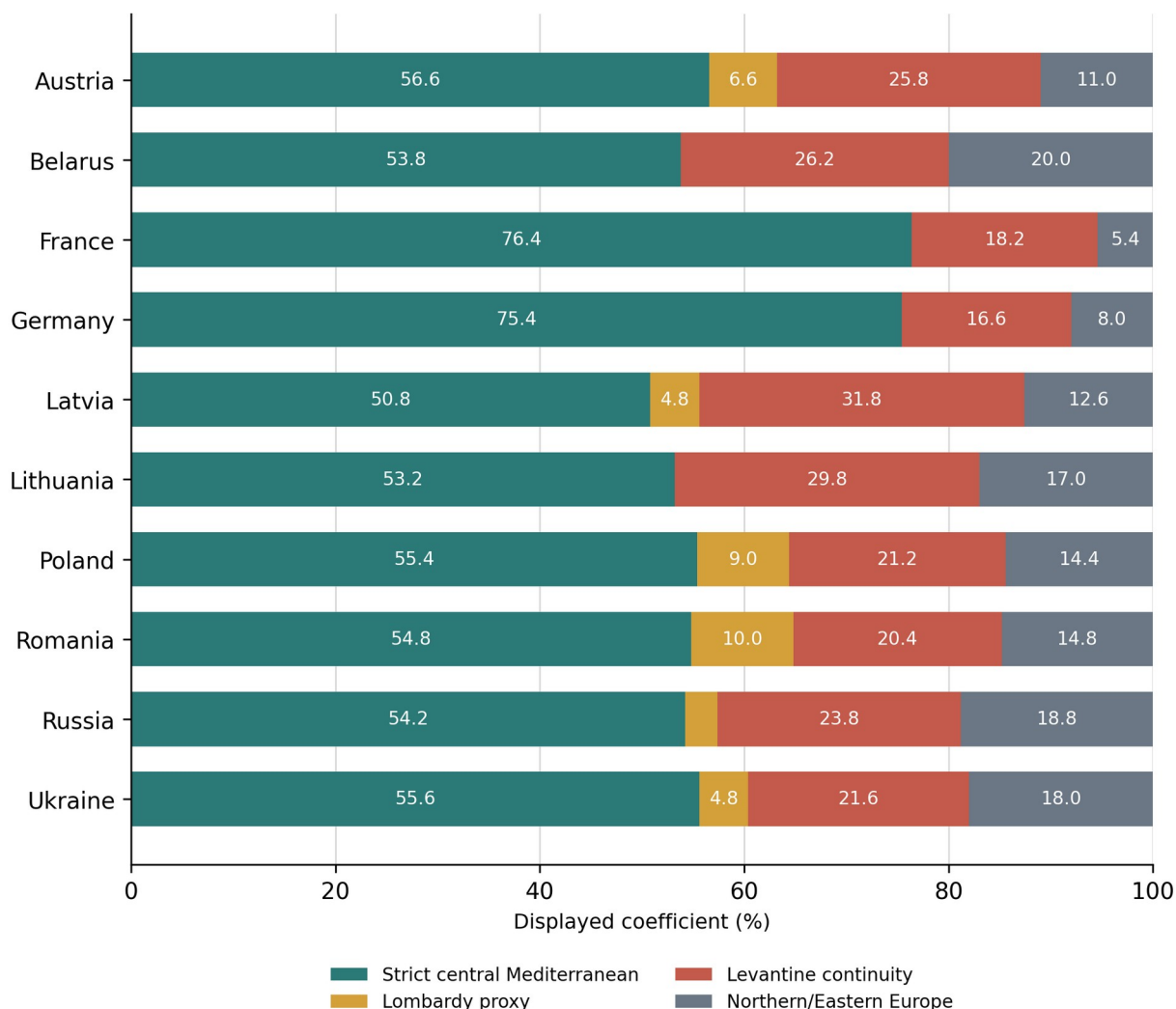


Figure 6. Experiment 14 regional Ashkenazi stability. The strict central Mediterranean block is a majority in all ten regional targets. Lombardy is shown separately and is not classified as Southern Italian. Levantine continuity and Northern and Eastern European sources remain secondary in every region. The regional averages are unequal in sample size and the coefficients are descriptive panel allocations. For the German target, the 16.6 Levantine-continuity and 8.0 Northern/Eastern European blocks generate strongly opposing twenty-five-dimensional displacements around the central Mediterranean centroid. Their weighted magnitudes are nearly equal, so the category totals should not be read as independent additive ancestry fractions.

Geometric Interpretation of Opposing Proxy Weights in Experiment 14

Experiment 14 also clarifies why an Ashkenazi target can remain close to Southern Italian and Italian Jewish populations while receiving nonzero Levantine and Northern/Eastern European coefficients. For Ashkenazi Germany, the displayed solution is 75.4 strict central Mediterranean, 16.6 Levantine continuity, and 8.0 Northern/Eastern European. These values are not three independent historical layers added sequentially to a Southern Italian baseline. They are weights in a constrained reconstruction of one twenty-five-dimensional target.

Using coefficient-weighted centroids for the sources retained in the German solution, the Levantine and Northern/Eastern European displacement vectors from the central Mediterranean centroid have a cosine similarity of -0.786. They therefore point in strongly opposing directions. Their weighted magnitudes are also nearly equal: 0.01024 for the 16.6 Levantine-continuity block and 0.01098 for the 8.0 Northern/Eastern European block. The smaller northern coefficient produces a comparable displacement because its centroid lies farther from the central Mediterranean centroid. Combining both shifts leaves a displacement of only 0.00697 from the central Mediterranean centroid.

A PCA calculated from all Experiment 14 sources and targets shows the same directional pattern. Along PC1, which accounts for 87.54 percent of variance in this declared coordinate set, the Levantine shift is -0.00869 and the Northern/Eastern European shift is +0.01083, leaving a net shift of +0.00214. The sign of a PCA axis is arbitrary; the relevant result is that the shifts oppose one another and nearly cancel. The full-dimensional calculation, rather than the projection, is the primary evidence.

Direct distances provide a complementary constraint. Ashkenazi Germany is 0.01470 from Reggio Calabria, 0.01588 from Italki, 0.01801 from Maltese, and 0.01978 from western Sicily. The corresponding distances are 0.05331 to Druze, 0.05875 to Lebanese Arab Christians, 0.06385 to Palestinian Arab Christians, and 0.07521 to Samaritans. These distances do not identify a unique ancestral population, but they show that the German target's overall position is much closer to the central Mediterranean references than to the retained Levantine-continuity proxies.

The 16.6 Levantine-continuity coefficient therefore should not be interpreted as evidence that a discrete 16.6 percent Levantine or Israelite layer was added to an otherwise Southern Italian population. In this model, the Levantine and northern sources partly function as opposing correction vectors around a predominantly central Mediterranean position. This does not establish the complete absence of Levant-related ancestry; it constrains the historical meaning that can be assigned to the displayed coefficient.

Ancient Levant Positive Controls: Experiments 9 and 15

Experiment 9 is a full twenty-five-dimensional distance control. Ancient Levantine targets are represented overwhelmingly by modern Samaritan and Palestinian Arab Christian controls. Peki'in, Ain Ghazal, and Middle Bronze Age Megiddo are modeled as 100.0 Samaritan in the declared competition; later Bronze and Iron Age targets remain majority Levantine. Central Mediterranean coefficients are nearly absent, averaging 0.1 for Rhodes and 0.8 for Calabria, with only Iron Age III Beirut showing a small 1.4 Rhodes and 9.8 Calabria signal. The method therefore does not automatically assign central Mediterranean sources to every eastern Mediterranean target.

Experiment 15 provides the stronger temporal continuity control by modeling three Iron Age coastal southern-Levantine averages against pre-Iron-Age Levantine site-period averages plus transparent Mycenaean and Anatolian Early Bronze Age composites. Philistine Ashkelon is allocated 100.0 to earlier southern-Levantine sources, Achziv 95.4, and Beirut 74.0. The unweighted target average is 89.8 pre-Iron-Age southern Levant, 7.3 Mycenaean composite, and 2.9 Anatolian composite, at average distance 0.01805080. The result does not exclude all external ancestry in these populations; it demonstrates that the framework recovers overwhelming local Levantine continuity when appropriate earlier references are supplied.

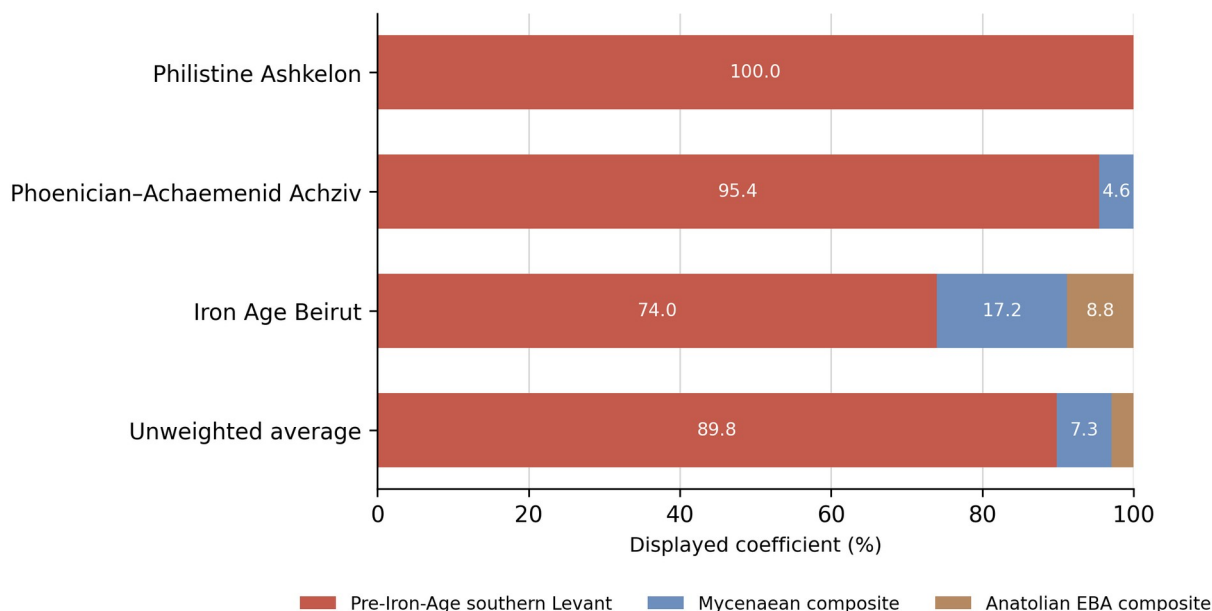


Figure 7. Experiment 15 ancient Levant positive control. Pre-Iron-Age southern-Levantine sources account for 100.0 in Philistine Ashkelon, 95.4 in Achziv, and 74.0 in Beirut, with an unweighted average of 89.8. The Mycenaean and Anatolian composites remain secondary and are not forced into every target. Values are constrained similarity weights among archaeological population averages, not literal genealogical percentages.

Integrated Interpretation

The fifteen experiments form a connected sensitivity suite. Prehistoric controls show why one Natufian coefficient cannot define historical origin. Modern full panels select Italki as the largest individual source. Matched omissions then transfer the missing signal first to related Mediterranean populations and, when that wider space is removed, to an inflated combination of Lombardy and Levantine proxies. The concurrent 47.18% deterioration in fit is critical: the stronger Europe–Levant cline appears because relevant comparators were removed, not because the reduced panel explains the target better.

Reciprocal Italki models show that the shared central Mediterranean direction is present in an Italian Jewish target without requiring later Ashkenazi residence in the Rhineland or Eastern Europe. Regional experiments show that the pattern is not confined to one pooled subgroup, and ancient positive controls show that the framework recovers strong Levantine continuity when genuine ancient Levantine targets are tested. Together, these controls weigh against simple explanations based solely on later northern host admixture or on an intrinsically optimal Northern Italian-plus-Levantine source pair.

Within the limits of constrained Global25 modeling, the repeated panel behavior supports a primary Italian Jewish and central Mediterranean-associated structure shared by Italki and Ashkenazi Jews. The smaller Levantine-continuity and Northern/Eastern European allocations should not automatically be treated as independent historical layers. In the German regional model, their full-dimensional vectors point in strongly opposing directions and have nearly equal weighted magnitudes, leaving little net displacement from the central Mediterranean position. The exact coefficients are therefore model-dependent proxy weights rather than historical percentages. Evidentiary weight comes from matched fit changes, category-level redistribution, reciprocal controls, regional targets, vector geometry, ancient positive controls, and convergence with independent qpAdm and FST analyses.

Part X. Italian Jews as a Falsification Test of Northern Italian and Convergent Origin Models

A broad consensus exists across population genetic studies that Ashkenazi Jews derive a substantial proportion of their autosomal ancestry from European sources. Across two-way and three-way admixture models, this European component is commonly estimated at approximately fifty percent, with the remaining ancestry attributed primarily to Levantine or eastern Mediterranean proxies such as Lebanese populations. The presence of a large Italian-associated component in Ashkenazi Jews is therefore not disputed. The point of disagreement concerns the geographic attribution of that Italian ancestry.

In most published models, the European component in Ashkenazi Jews is assigned to Northern or Central Italian proxies, often in combination with Levantine references, while Southern Italian populations are minimized, excluded, or treated as unsuitable stand-ins. This practice persists despite repeated observations that Ashkenazi Jews cluster most closely with Southern Italians in Mediterranean-inclusive autosomal PCA space and exhibit their lowest autosomal FST distances to Southern Italian and Aegean populations. The continued reliance on Northern Italian proxies therefore reflects reference population design rather than constraints imposed by the data.

Italian Jews provide a useful comparative constraint because long-established Jewish communities existed in Italy before the documented formation of Ashkenazi communities north of the Alps. Their genetic position can therefore test whether an Italian-associated component necessarily resembles Northern Italian populations. This comparison is informative but not decisive by itself, because Italian Jewish demographic history also includes drift, mobility, and admixture.

In the autosomal comparisons reviewed here, Italian Jews and Ashkenazi Jews occupy nearby Southern Italian and central Mediterranean space, whereas Northern Italian comparators are more distant in the tested panels. Founder drift can alter within-cluster position and allele frequencies, so the relevant observation is not perfect overlap but the repeated regional ordering across PCA, FST, and reciprocal modeling.

Independent genome-wide analyses further corroborate this interpretation. Campbell et al. (2012) analyzed a broad panel of Jewish and non-Jewish populations using autosomal PCA, neighbor-joining trees, FST distances, and identity-by-descent sharing. In that study, Italian Jews cluster directly adjacent to Ashkenazi Jews and form a shared European and Mediterranean Jewish branch together with Greek and Turkish Sephardic Jews. Northern Italian populations, included explicitly as comparators, instead cluster with European non-Jewish populations and are clearly separated from the Jewish branch across all analytical frameworks. Italian Jews do not occupy an intermediate position between Italians and Middle Eastern Jews, nor do they show reduced Jewish–Jewish identity-by-descent sharing. Rather, they participate fully in the shared Jewish ancestry network, indicating long-term regional continuity rather than secondary admixture with surrounding Italian populations. This structure is incompatible with models that assign Ashkenazi Jews substantial Northern Italian ancestry while treating Italian Jews as marginal or derivative.

The same structural limitation is visible in influential secondary syntheses. A central example is the neighbor-joining tree presented by Ostrer and Skorecki (2013), which has been widely cited as a summary of autosomal relationships among Jewish diaspora populations and surrounding non-Jewish groups. In this tree, Italian Jews cluster adjacent to Ashkenazi Jews within a Southern European and central Mediterranean branch. However, the European reference set used in this analysis is dominated by Northern and Central European populations and a limited set of Italian proxies, while Southern Italian and Sicilian populations are omitted despite their direct relevance to Mediterranean population structure and historically plausible source populations. Under these conditions, apparent intermediacy can be produced by reference omission rather than reflecting true ancestry.

The reciprocal tests in Part IX provide an author-generated falsification of a Northern Italian or later Northern European explanation for Italian Jewish placement. In Experiment 6, Italki resolves entirely through displayed Maltese, Rhodian, Sicilian, and Calabrian coefficients, with no Northern Italian, Tuscan, or Northern and Eastern European source retained. Experiment 13 expands the competition by adding modern Levantine-continuity controls while excluding Ashkenazi Jews as a source.

In Experiment 13, the Italki target resolves as 70.6 percent strict central Mediterranean, 74.8 percent broader Italian and central Mediterranean when the 4.2 Lombardy coefficient is included, 25.2 percent Levantine continuity, and 0.0 percent Northern and Eastern European in the displayed solution. Because German, Polish, and Russian sources were allowed to compete but were not retained, the shared Italki–Ashkenazi central Mediterranean placement cannot be attributed to later Rhineland or Eastern European admixture. These values remain panel-dependent similarity weights, and the reciprocal structure is more informative than any one modern proxy coefficient.

Taken together, these findings constrain models that assign the Ashkenazi Italian-associated component mainly to Northern Italy while treating Italian Jews as unrelated to that ancestry. Such models must explain why Italian Jewish and Ashkenazi targets repeatedly share a central Mediterranean direction, why Northern and Eastern European sources are unnecessary in the reciprocal Italki model, and why removal of central Mediterranean comparators inflates both Lombardy and Levantine coefficients while worsening fit. A shared older Mediterranean-associated substrate is a more parsimonious interpretation of these observations, although the present analyses cannot identify a single historical community or admixture date.

Italian Jews therefore function as a comparative falsification test of a narrow proxy claim, not as proof of one exclusive origin scenario. Their autosomal and reciprocal-model placement is more consistent with a Southern Italian and adjacent Aegean-associated component than with a specifically Northern Italian or later Northern European source. Ashkenazi-specific founder effects and later Eastern European admixture can then modify that older structure without defining its primary geographic affinity.

Part XI. Conclusions

This study tested whether the familiar Ashkenazi Europe-Levant cline is stable to the inclusion of historically and genetically relevant central Mediterranean comparators. It is not stable in the matched experiments. With Italki, Southern Italian, Sicilian, Maltese, and Aegean sources present, they form the largest fitted direction; when they are removed, north-shifted Italian and Levantine proxies absorb the missing signal.

The mechanism is transparent. A constrained model can represent only the source directions it is given. If the central Mediterranean region nearest the target is omitted, the remaining European and Levantine sources can bracket that missing space and produce an apparently informative cline. In Experiments 10-12, this is not a verbal possibility: removal of the wider central Mediterranean panel worsens distance by 47.18%, raises Lombardy from 9.8 to 38.0, and raises Levantine continuity from 22.8 to 52.0. The stronger two-pole mixture is therefore accompanied by a substantially poorer fit.

This result does not imply that Levant-related ancestry is absent. It shows that a nonzero Levantine-proxy coefficient may partly encode a geometric correction rather than a discrete additive ancestry layer. In the German regional model, the 16.6 Levantine-continuity and 8.0 Northern/Eastern European blocks have nearly equal weighted magnitudes in strongly opposing directions, leaving the target close to the Southern Italian-Italki region. Reciprocal Italki models, ten regional Ashkenazi targets, qpAdm and FST comparisons, and Mediterranean-inclusive published analyses support the same regional ordering. Together, they support a dominant Italian Jewish and central Mediterranean structure, while the residual proxy coefficients remain model-dependent and cannot be converted directly into historical ancestry percentages.

Y-chromosomal frequencies are compatible with this interpretation but are not decisive. J2, J1, and E lineages span several Mediterranean regions, and founder effects can substantially change their frequencies. Their main contribution here is to show that coarse haplogroup labels should not be assigned a uniquely Levantine origin when relevant Southern European comparators are absent.

The CCR5-Δ32 distribution and other founder phenomena provide ancillary evidence for European input and strong post-admixture drift, but they do not quantify population-wide ancestry. The principal inference remains the controlled response of genome-wide models to comparator inclusion and omission.

Historical evidence makes Southern Italy, Sicily, and the wider central Mediterranean plausible contexts for the older component identified by the genetic comparisons. It does not identify a unique source population or date. Future work should test these alternatives with raw-genotype analyses, denser sampling of Southern Italian and Italian Jewish populations, and ancient genomes from relevant Mediterranean Jewish communities.

The central conclusion is methodological and demographic: omission of central Mediterranean comparators can create or exaggerate an Ashkenazi Europe-Levant cline, while their inclusion yields better-fitting models dominated by Italian Jewish and central Mediterranean-associated affinity. This does not erase eastern Mediterranean or Levantine continuity; it changes its scale and interpretation. Any model of Ashkenazi origins should therefore report explicit panel-sensitivity tests before treating clinal placement or proxy coefficients as historical ancestry proportions.

Supplementary Materials

Supplementary Tables S1–S4 present Y-chromosomal frequencies and regional comparisons; Tables S5–S7 summarize autosomal estimates, published FST proximity results, and CCR5-Δ32 frequencies; Table S8 presents the author-generated qpAdm models; Table S9 presents author-generated autosomal FST distances; and Table S10 presents FST calibration benchmarks. The public repository cited at the end of Methods contains the complete Global25 Experiments 1–15, including declared inputs, raw and grouped outputs, comparison tables, figures, screenshots, scripts, methods documentation, and checksums. The versioned archival release is <https://doi.org/10.5281/zenodo.21383280>.

Ethics Statement

This study reports secondary analyses of previously published or publicly distributed population-level data and introduces no new participant recruitment, biological sampling, or sequencing.

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Competing Interests

The author declares no competing interests.

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